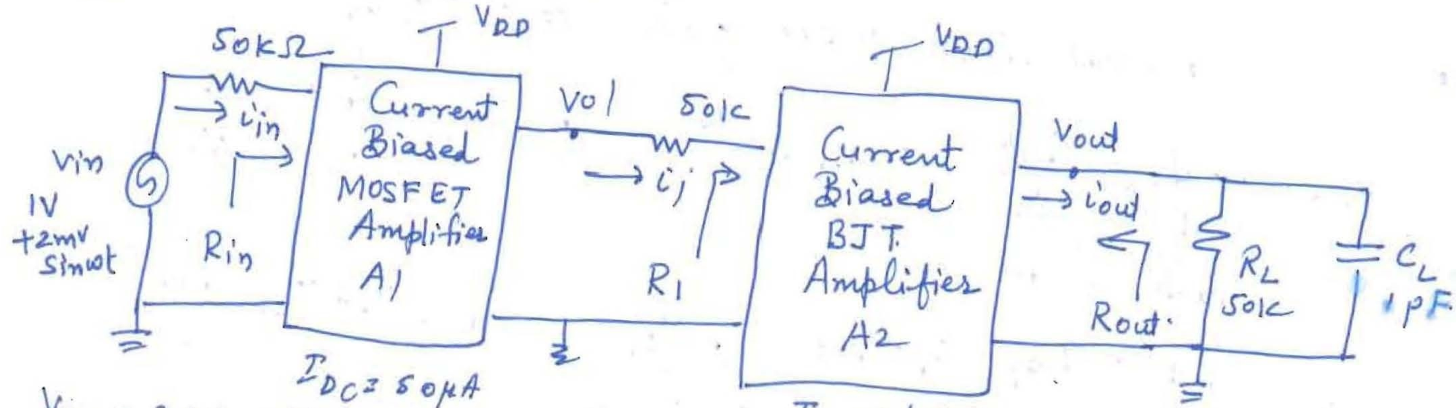
Part A CB

Q.

Sol



$V_{DD} = 3.3V = V_{CC}$.

$\mu_n C_{ox} = 140\ \mu A/V^2$ $V_T = 0.7V$, $\lambda = 0.1V^{-1}$
$\mu_p C_{ox} = 40\ \mu A/V^2$ $V_T = -0.7V$ $\lambda = 0.1V^{-1}$ $\Big\}$ $V_{ov} = 0.2V$

$\beta = 100$, $V_{CE,sat} = 0.2V$, $V_A = 100V$, $\frac{kT}{q} = 25mV$, $I_S = 10^{-14}A$

All Devices Biased in active region

(a) MOSFET $g_m = \frac{2I}{V_{OV}} = \frac{2 \times 50\ \mu A}{0.2} = 0.5\frac{mA}{V}$, $r_0 = \frac{1}{\lambda I} = \frac{1}{0.1 \times 50\ \mu A} = 200\ k\Omega$

BJT $g_m = \frac{I_c}{V_t} = \frac{100\ \mu A}{25mV} = 4\frac{mA}{V}$, $r_0 = \frac{V_A}{I_c} = \frac{100V}{100\mu A} = 1M\Omega$, $r_\pi = \frac{\cancel{V_{BE}}\ V_T}{I_c}\beta$

$V_{BE} = V_{Th} \ln\left(\frac{I_c}{I_s}\right) = 25mV \ln\left[\frac{50\ \mu A}{10^{-14}}\right] = 25mV \ln\left[50 \times 10^8\right]$
$= 0.558$

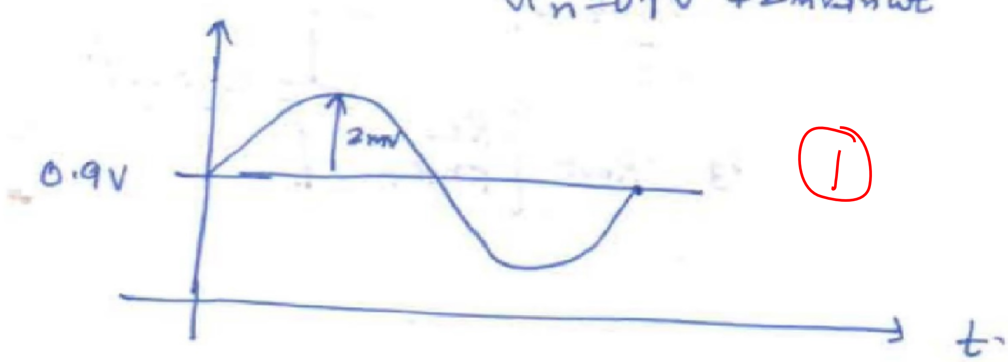
So, $r_\pi = \frac{\cancel{0.558}\ 0.025 \times 100}{100\ \mu A} = \frac{\cancel{55.8}\ 2.5}{\cancel{100}} \times 10^6 = \cancel{558}\ 25\ k\Omega$

5 ~~3~~

(b)

Input Wave form w.r.t time.

$V_{in} = 0.9V + 2mV \sin\omega t$



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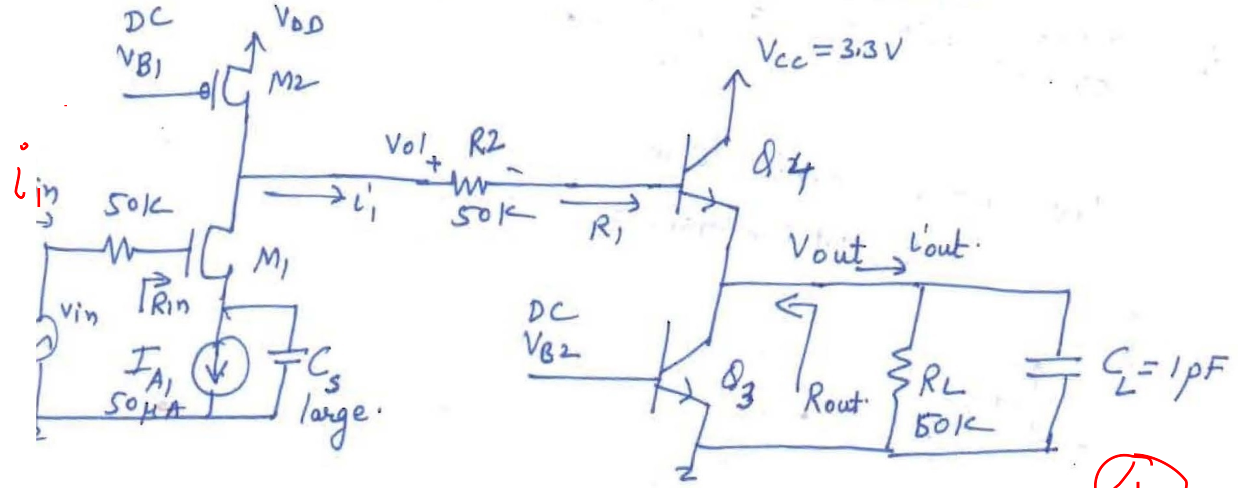
(c) $A_1 \rightarrow$ CSA, A_2 = Emitter Follower



~~3~~ 3

4

(e) low frequency small signal voltage gain $A_v = V_{out}/V_{in}$ using $G_m R_{out}$

$$\frac{V_{out}}{V_{in}} = \underbrace{\frac{V_{out}}{i_{out}}}_{R_{out}} \times \underbrace{\frac{i_{out}}{V_{in}}}_{G_m} = G_M R_{out} \quad ; \quad \boxed{G_m = \frac{i_{out}}{v_{o1}} \times \frac{v_{o1}}{v_{in}}}$$

$$i_{out} = g_{m4}(V_{o1} - v_{out}), \quad V_{o1} = -g_{m1}(r_{o1} \| r_{o2})\, v_{in}$$

$$v_{out} = i_{out}\,(r_{o4} \| r_{o3})$$

$$\Rightarrow \quad i_{out} = g_{m4}\left(V_{o1} - i_{out}\, r_{o4}/2\right)$$

9

$$i_{out}\left[1 + \frac{g_{m4} r_{o4}}{2}\right] = g_{m4} V_{o1}$$

$$\frac{i_{out}}{V_{o1}} = \frac{g_{m4}}{1 + g_{m4} r_{o4}/2}$$

2

$$\text{Now } G_m = \frac{g_{m4}}{1 + \frac{g_{m4} r_{o4}}{2}} \times -g_{m1}(r_{o1}/2) = \frac{-g_{m1} g_{m4}(r_{o1}/2)}{\left(1 + \frac{g_{m4} r_{o4}}{2}\right)}$$

$$= \frac{0.5\,mA/V \times 4\,mA/V\,(0.1\,M)}{(1 + 4m \times 1M/2)} = \frac{0.2}{1500}\ A/V$$

(d)

$$R_1 = r_\pi + (\beta+1)(r_{o3} \| R_L)$$

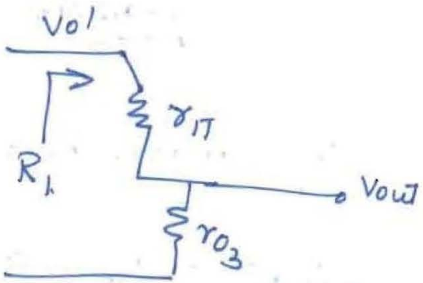
$$\cong \cancel{558}\,25K + (100)\,\cancel{1M}\ 0.5M$$

$$\boxed{R_1 \cong \cancel{100}\,M\Omega}$$

$50 M\Omega$

~~2~~

1



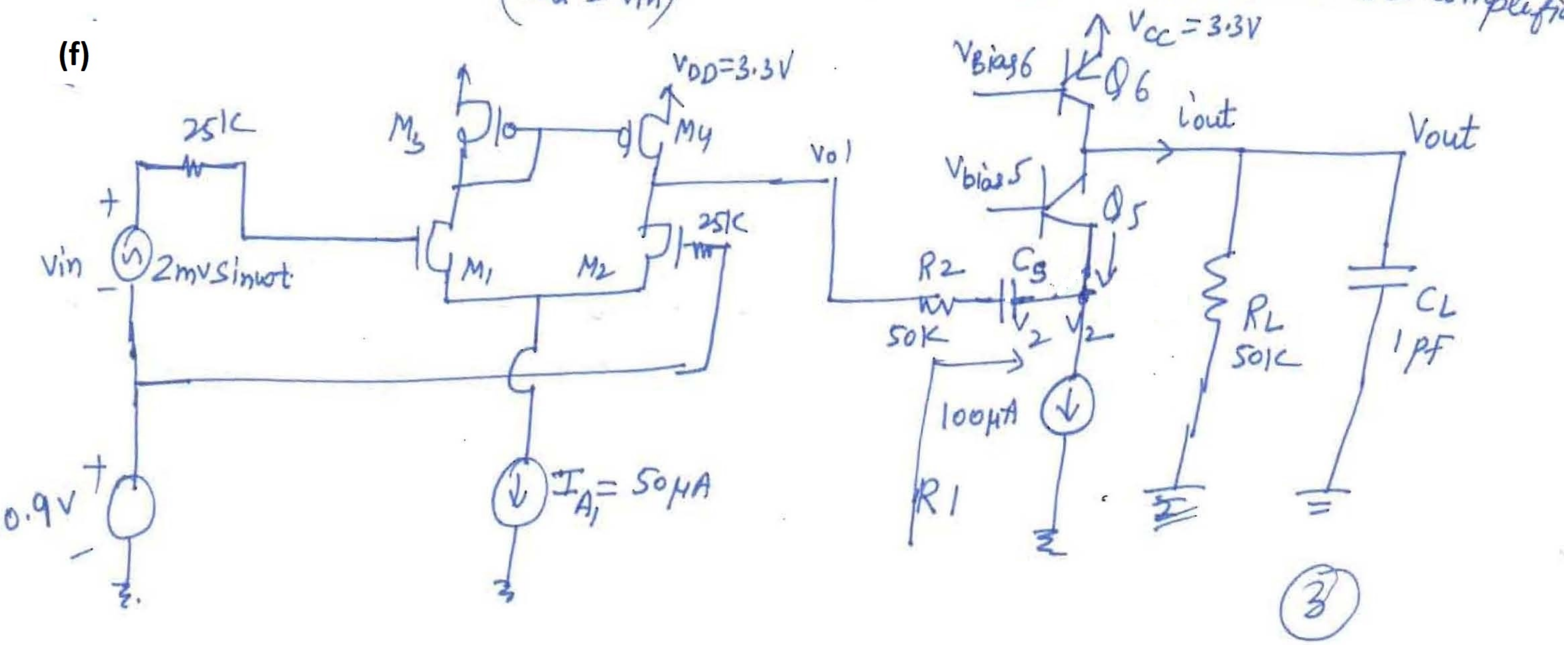
DC power dissipated = Vdd (50+100)uA
= 0.495 mW

with Iref ⇒ DC Power = 0.98 mW

1

A_1 is Single output differential amplifier & A_2 is Common base amplifie
$(V'_{id} = V_{in})$

(f)



(g)

$$A_{cm} = \frac{V_{out}}{V'_{icm}} = \frac{1}{1+2g_m R_{ss}} = 0.01$$ 1

Rss= 100 K, gm= 0.5 mA/V

for perfect matching (

$$|A_{dm}| = g_{m_1}\,(r_{o_4} \| r_{o_2} \| 1/g_{m_5})$$

$g_{m_5} = 4\,\frac{mA}{V}$

$$= \frac{1}{2}\frac{mA}{V} \times \cancel{0.1M\Omega}\ 250$$

$$= \frac{1}{2} mA \times \cancel{100K}\ 250$$

$$= \cancel{50}\ 125$$ 1

(h)

$$|V_{offset}| = \frac{V_{o1}}{A_{dm}} = \frac{V_{o1}}{\cancel{50}\ 125} = \cancel{\frac{0.1m}{50} = 0.002\,m = 2\mu V}$$

0.83 mV

1

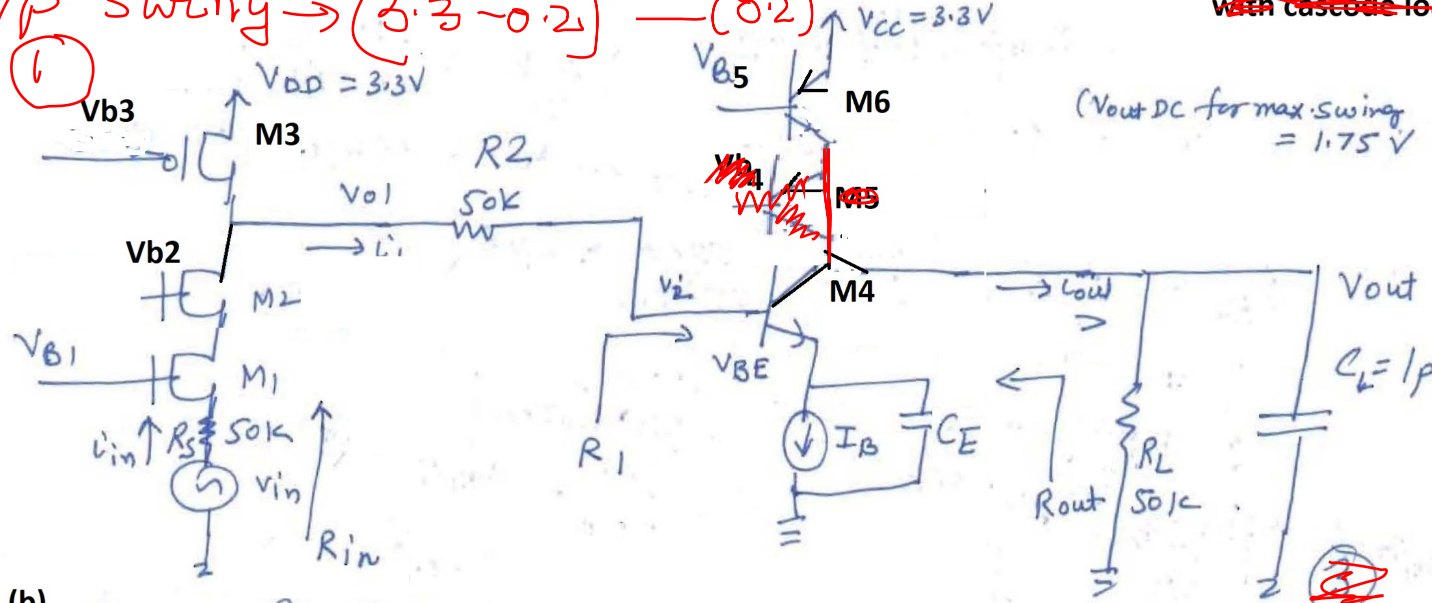
Q2

(a) stage-1 cascode Common Gate amplifier, Stage-2 Common Emitter Amp. ~~with cascode load~~

2 2

O/P Swing → (3.3 − 0.2) — (0.2)

1



(b)

$$R_{in,\,CGA} = \frac{R_L + r_{o1}}{1 + g_{m1} r_{o1}} = \frac{r_{o2} + r_{o1}}{1 + g_{m1} r_{o1}} = \frac{(0.2 + 0.2)\,M\Omega}{1 + 500\mu \times 0.2M} = \frac{0.4\,M}{101} = 3.96k\Omega \simeq 4k\Omega$$

2

$$= 2/g_m \simeq 4K$$

(c)

$$V_{out} = i'_{out}(50k \,\|\, 100k) \approx i'_{out}\, 50k = \cancel{-50\,mV} \;\; 150\,mV$$

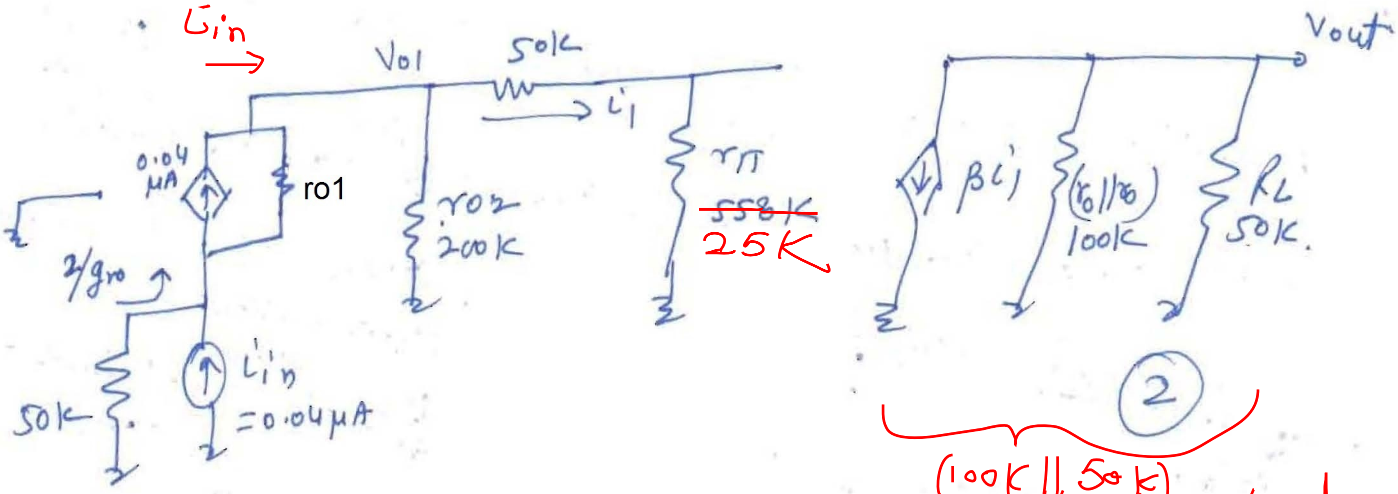
$$i'_{out} = -\beta\, i'_1 = \cancel{-1\,\mu A} \qquad -100 \times 0.029 = 2.9\,\mu A \; ; \quad \beta = 100$$

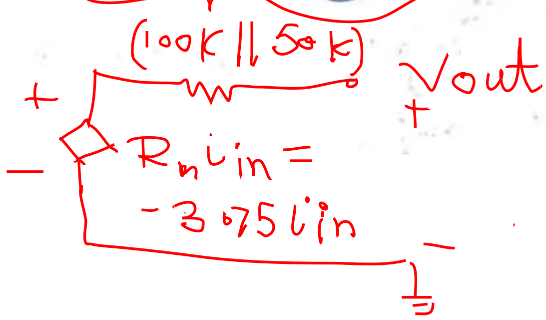
$$i'_1 = \frac{200k}{200k + (50k + \cancel{558k}\; 25k)}\, i'_{in} = \cancel{0.01\,\mu A} \;\; 0.73\,\mu A \times 0.04 = 0.029$$

$$i'_{in} = \frac{V_{in}}{R_s} = 0.04\,\mu A$$

$$\Rightarrow R_M = \frac{V_{out}}{i'_{in}} = \frac{-150\,mV}{0.04\,\mu A} = \cancel{-1250}\; -3750 \times 10^{-3}$$

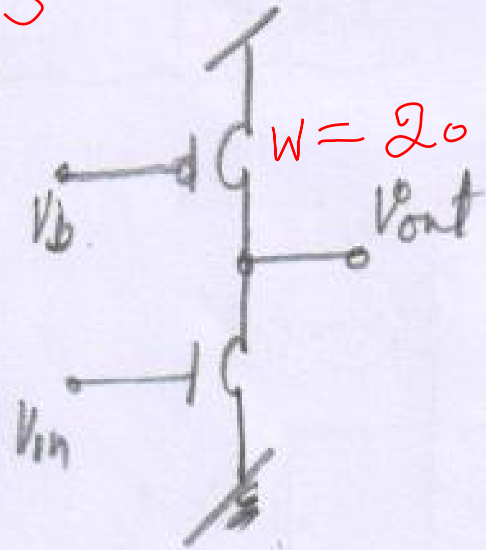
$$R_M = \cancel{3.75}\,\Omega = -3.75\,\Omega$$



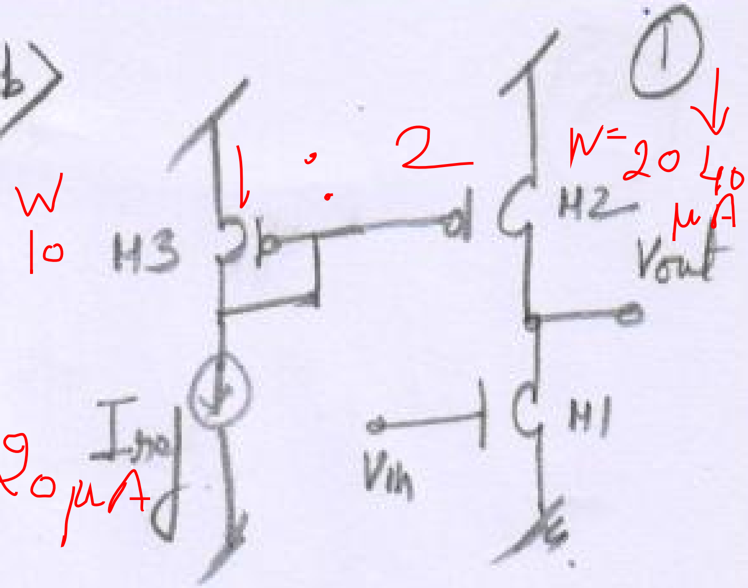


Q 3

a)



b)

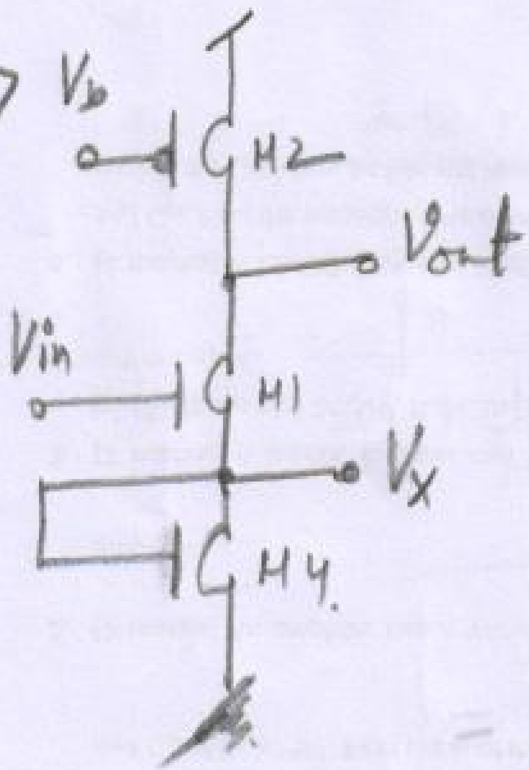


$$I_{out} = \frac{W_2}{W_3} \times I_{ref}$$

$$W_3 = 20\mu A \times \frac{20\mu A}{40\mu A}$$

$$W_3 = \cancel{5\mu A} \quad 10\mu m$$

c)



@ Node Vout

d) $A_V = \dfrac{-g_m R_D}{1 + g_m R_S}$

$R_D = r_{o2}$

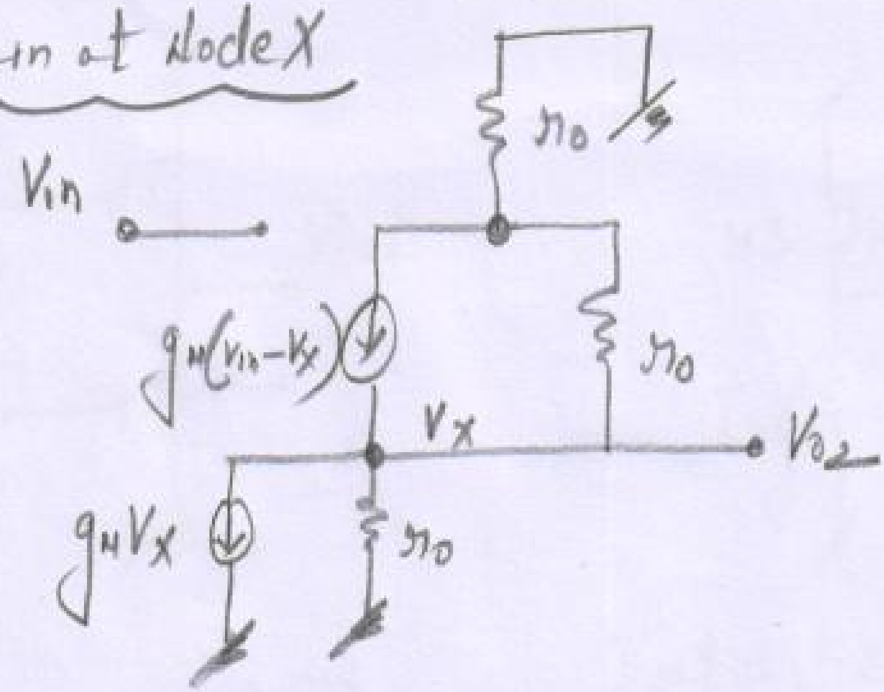
$R_S = \dfrac{1}{g_{m4}}$

- $g_m = \dfrac{2 I_D}{V_{ov}} = \dfrac{40\mu A}{0.2} = 200\mu A/V$
- $r_o = \dfrac{1}{\lambda I_D} = \dfrac{1}{0.1 \times 40\mu A} = 250k.$
- $g_{m_4} = 200\mu A/V$
- $1/g_{m_4} = \dfrac{1}{200\mu A} \cong 5k\Omega$

d) $A_V = \dfrac{200\times10^{-6} \times 250\times10^{3}}{1 + 200\times10^{-6} \times 5\times10^{3}}$

$A_V = \dfrac{250}{5} \approx 50$

e) Gain at Node X



(2)

$$g_m(v_{in}-V_x) = g_m V_x + \frac{V_x}{r_0} + \frac{V_x}{2r_0}$$

$$g_m V_{in} = 2g_m V_x + \frac{V_x}{r_0} + \frac{V_x}{2r_0} = V_x\left[2g_m + \frac{1}{r_0} + \frac{1}{2r_0}\right]$$

$$\frac{V_{o2}}{V_{in}} = \frac{V_x}{V_{in}} = \frac{g_m}{2g_m + \frac{1}{r_0} + \frac{1}{2r_0}} \approx \frac{g_m}{2g_m} = \frac{1}{2}$$

$$\boxed{\frac{V_{o2}}{V_{in}} = \frac{1}{2}}$$

f) No effect on CTR.

Q4

a) $V_o = -g_m V_i (r_o \| R_L) \Rightarrow$ Voltage gain: $\frac{V_o}{V_i} = -g_m(r_o \| R_L)$

[4]

given $r_o = R_L = 1/g_m \Rightarrow r_o \| r_L = 1/2g_m$

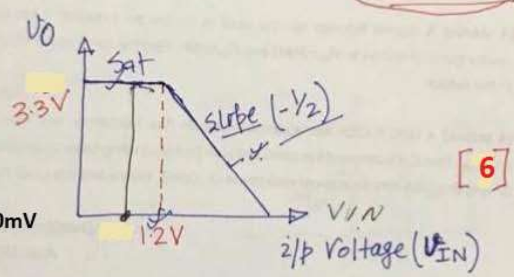
$$\frac{V_o}{V_i} = -g_m \cdot \frac{1}{2g_m} = -\frac{1}{2}$$

$$\text{gain in dB} = 20 \log \left| \frac{V_o}{V_{in}} \right| = 20 \log \frac{1}{2}$$

$$= -6.02 \text{ dB}$$

[4]

b) VTC



Vo, peak= 200/2 = 100mV

Vin peak=200mV

c) $P_L = \frac{V_{o\,rms}^2}{R_L} = \frac{(0.1/\sqrt{2})^2}{1000} = 0.005 \text{ mW}$

$P_{dc} = 3.3 \times 1 \text{ mW} = 3.3 \text{ mW}$

$$\text{Power efficiency } (\eta) = \frac{0.005}{3.3} \times 100\%$$

$$= \cancel{0.167\%} \quad 0.152 \pm 0.001\%$$

[5]

Part B (OB)

① Sol^n:

$A_{OL} = 100\,dB$ $\quad A_{OL} = 10^5$

$A_{CL} = 40\,dB$ $\quad A_{CL} = 10^2$

$A_{CL} = \dfrac{A_{OL}}{1 + A_{OL}\beta}$ $\quad A_{OL} \cdot \beta = 10^3$

$\beta = 10^{-2}$

So β (feedback factor) $= 10^{-2} \rightarrow$ (2)

$A_{OL}\beta$ (Loop gain) $= 10^3 \rightarrow$ (2)

Shunt $\rightarrow$ $R_{if} = \dfrac{R_{in}}{(1 + A_{OL}\beta)} = 100\,\Omega$ (shunt Resistance ↓) (2)

$R_{in} = 100\,k\Omega$

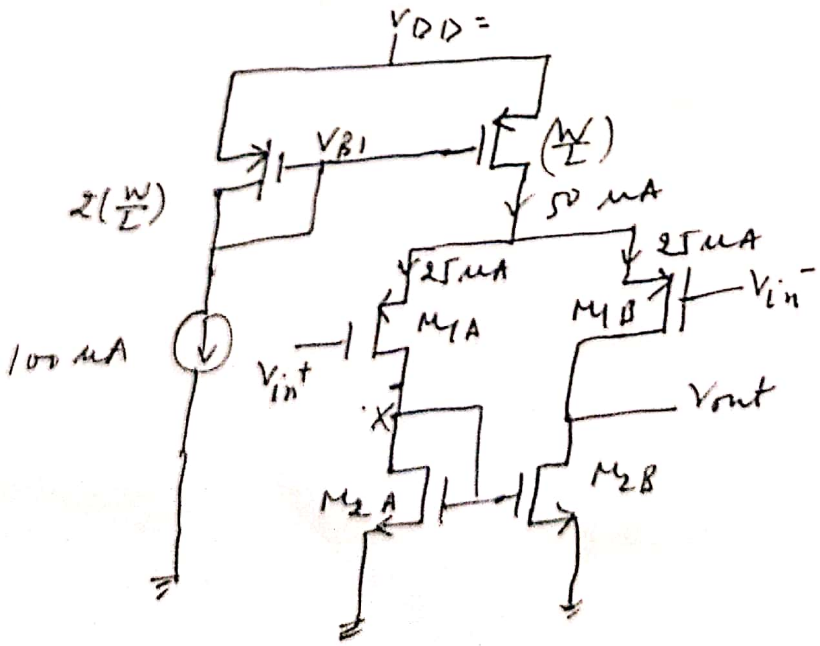
$R_{out} = 0.1\,k\Omega$

Series $\rightarrow$ $R_{of} = R_{out}(1 + A_{OL}\beta) = 0.1\,M\Omega = 100\,k\Omega$ (2)

3-dB Bandwidth = BW increases by $(1 + A_{OL}\beta)$

So $BW' = 1\,K$ rad/sec $(1 + A_{OL}\beta) = 10^6$ rad/sec (2)

②



$V_{OV} = 0.25\,V$

$V_A |_{n,p} = 100\,V$

$V_{TH} |_{n,p} = 0.65\,V$

Sol^n: from current mirror property:
current passing through M_{1A} & M_{1B} will be $25\mu A$ respectively.

$$g_m = \frac{2I_D}{V_{OV}} = \frac{2\times 25\times 10^{-6}}{0.25} = 0.2\, mA/V$$

$$r_o|_{n,p} = \frac{V_A}{I_D} = \frac{100\,V}{25\times 10^{-6}} = 4\,M\Omega$$

a) Differential gain $(A_{DM}) = -g_{m_p}(r_o|_n \parallel r_o|_p)$

$$= -0.2\times 10^{-3}[4 \parallel 4]\times 10^{6}$$

$[\,A_{DM} = -400\, V/V\,]$ — 2

b) V_{out} (Swing) :

$V_{out}(max) = V_{DD} - 2V_{OV}$

$= 2.5 - 0.5 = 2\,V$

$V_{out}(min) = V_{OV} = 0.25\,V$

So V_{out} (Swing) $= 2 - 0.25 = 1.75\,V$ — 2

c) Value of ICMR : $V_{out}(DC) = V_{Th} + V_{ov} = 0.90\,V$

V_{in} (lower limit) $= V_{out}(DC) + V_{Th} = 0.9 + 0.65 = \boxed{1.55\,V}$

V_{in} (Upper limit) $= V_{DD} - V_{ov} + V_{GS}|_P = 2.25 + 0.90$

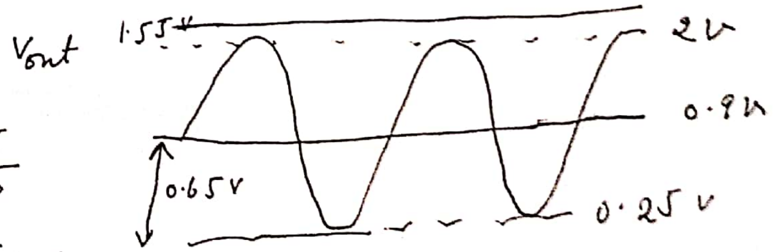
$= \boxed{3.15\,V}$

3

d) $V_{out} = V_{out}(DC) - 0.1\sin\omega t \times 400$ mV

$V_{out} = (900 - 40\sin\omega t)$ mV — 3

e)



So $V_P|_{max} = \frac{0.65}{A_{DM}} = \frac{0.65}{400}$

$= 1.625$ mV — 3

Scanned with CamScanner

f) Input offset voltage

$$= \frac{V_{ov}}{2}\left(\frac{\Delta(W/L)}{W/L}\right)$$

$\frac{\Delta(W/L)}{W/L} = 5\%$

$$= \frac{0.25}{2} \times 0.05$$

$$= \underline{6.25\,mV} \rightarrow 3$$

g) Dominant pole f $\frac{1}{R_{out} C_{Load}} = \frac{1}{2M\Omega \times 2pF}$

$C_x = 200\,fF$

$C_{Load} = 2\,pF$

$$= 0.25 \times 10^{6}$$

$$\underline{\omega_{\text{dominant pole}} = 250\,K\ rad/s} - 2$$

$$\underline{\text{Mirror pole frequency}} = \frac{g_{m_{In}}}{C_x} = \frac{0.2 \times 10^{-3}}{200 \times 10^{-15}} = 0.1 \times 10^{10}\ rad/sec$$

$$= \underline{1\ G\,rad/sec} \leftarrow 2$$

Part-B 6B

$$\frac{Y(s)}{X(s)} = \frac{2\times 20\left(\frac{j\omega}{20}+1\right)}{2\times 10\left[\frac{j\omega}{2}+1\right]\left[\frac{j\omega}{10}+1\right]} = \frac{20\left(\frac{j\omega}{20}+1\right)}{\left(\frac{j\omega}{2}+1\right)\left(\frac{j\omega}{10}+1\right)}$$

Unity Gain frequency:- $\left|\frac{Y(j\omega)}{X(j\omega)}\right| = 1.$

$$20\sqrt{\left(\frac{\omega}{20}\right)^2+1} = \sqrt{\left(\frac{\omega}{2}\right)^2+1}\sqrt{\left(\frac{\omega}{10}\right)^2+1}$$

$$400\left[\frac{\omega^2}{400}+1\right] = \left(\frac{\omega^4}{400}+\frac{104\omega^2}{400}+1\right)$$

$$\frac{\omega^4}{400}+\frac{(104-400)\omega^2}{400}-399=0$$

$$\omega^4 + -296\omega^2 - 399\times 400 = 0$$

$$\omega^2 = \underline{574.03} \text{ (or) } -278.03$$

$\boxed{\omega = 24 \text{ rad/sec}}$ Gain crossover freq.

Phase @ ω_{GCF} $\Rightarrow$ $\tan^{-1}\left(\frac{\omega}{20}\right) - \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{10}\right)$.

$= -102.4^\circ$ is the phase @ ω_{GCF}

Phase Margin $= 180^\circ - 102.4^\circ = \boxed{77.5^\circ}$

The system has one pole (20 rad/sec) and two zeros one each at 2 rad/sec and 10 rad/sec. Initial phase is 0° $(\omega=0)$. Therefore the system will never see a phase of 180°. $\therefore$ phase crossover frequency is undefined. Gain Margin is Infinity.

3dB Bandwidth:-

$$2\left[\frac{\omega^2}{400}+1\right] = \left[\frac{\omega^2}{4}+1\right]\left[\frac{\omega^2}{100}+1\right].$$

$$2\omega^2 + 800 = \omega^4 + 104\omega^2 + 400$$

$$\omega^4 + 102\omega^2 - 400 = 0 \qquad \omega^2 = 3.78 \text{ (or) } -105$$

3dB Bandwidth $= \boxed{1.94 \text{ rad/sec}}$

Since the pole at 10 rad/sec and zero at 20 rad/sec are two octaves above the pole at 2 rad/sec. We can say the pole at 2 rad/sec is the dominant pole.

$\therefore \omega_{3dB} = \boxed{2 \text{ rad/sec}}$

Gain of the CSA (M1-M2):- $-g_m(r_{o1} \parallel r_{o2})$

$= -0.2\,(400K \parallel 600K)$ mA/V

$\boxed{A_V = -48}$

This is also the gain of stage 2 CSA (M3-M4).

Cap. at node P:- $C_{gs1} + C_{gd1}(1+A_V)$

$= 100\,fF + 50(1+48)\,fF$

$\boxed{= 2.55\text{ pF}}$

Cap. at node-X :-

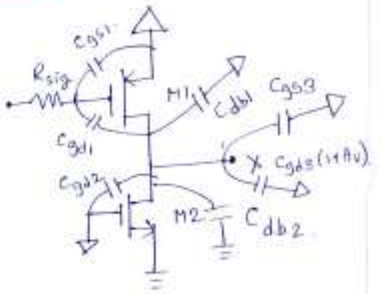
$$\underbrace{\left[C_{gd1} + C_{db1}\right]}_{\text{from M1}} + \underbrace{\left[C_{db2} + C_{gd2}\right]}_{\text{from M2}} + \underbrace{\left[C_{gs3} + C_{gd3}(1+A_v)\right]}_{\text{from M3}}$$

$$= 70fF + 80fF + 100fF + 50fF(1+48).$$

$$= \cancel{2.64pF} \quad \boxed{2.7pF}.$$

OCTC Method

When we combine all the intrinsic cap. that can be added.



$C_1 = C_{gs1}$; $100fF$

$C_2 = C_{gd1}$; $50fF$

$C_3 = C_{db1} + C_{gd2} + C_{db2} + C_{gs3} + C_{gd3}(1+A_v) = 2.65pF$

Therefore we will calculate three time constants

$R_1 = R_{sig} \| (R_{gs}) = 100k\Omega$

$R_3 = R_L' = r_{o1} \| r_{o3} = 240k\Omega$

$R_2 = R_{gd} = R_{sig}(1 + g_m R_L') + R_L' = 5.14\ M\Omega.$

$\tau_1 = R_1 C_1 = \boxed{10nS}$

$\tau_2 = R_2 C_2 = \boxed{13.621\mu S}$

$\tau_3 = \boxed{12nS}$

$\tau_H = \tau_1 + \tau_2 + \tau_3 = \boxed{13.643\mu S}$

$\omega_H = \boxed{73.29\ Krad/sec}$ (11.7 KHz).

Resistance at node-G :- $(r_{o2} \| r_{o4}) = 240k\Omega$

$$\omega_G \simeq (240k\Omega \times C'_{Load})^{-1} = 500\ Krad/sec.$$

$$C'_{Load} = 8.3\ pF$$

intrinsic cap. connected to node-G :-

$$C_{db4} + C_{gd4} + C_{gd3} + C_{db3} = 150fF \text{ (too small)}$$

$$\therefore C_{Load} \simeq \boxed{8.3pF}$$

$CMRR = \frac{A_{DM}}{A_{CM}}$ Given $A_{DM} = \frac{G_1}{(1 + \frac{j\omega}{P_1})}$ $P_1 = 40\ Mrad/sec$

We know A_{CM} has a zero at $\omega_z = (R_{SS}C_{SS})^{-1} = \boxed{100\ Mrad/sec.}$

$$CMRR = \frac{K \text{ (some constant)}}{\left(1 + \frac{j\omega}{40}\right)\left(1 + \frac{j\omega}{100}\right)} \qquad \boxed{\omega \text{ is in Mrad/sec}}$$

$$\omega_{3dB} = \boxed{35.28\ Mrad/sec}$$

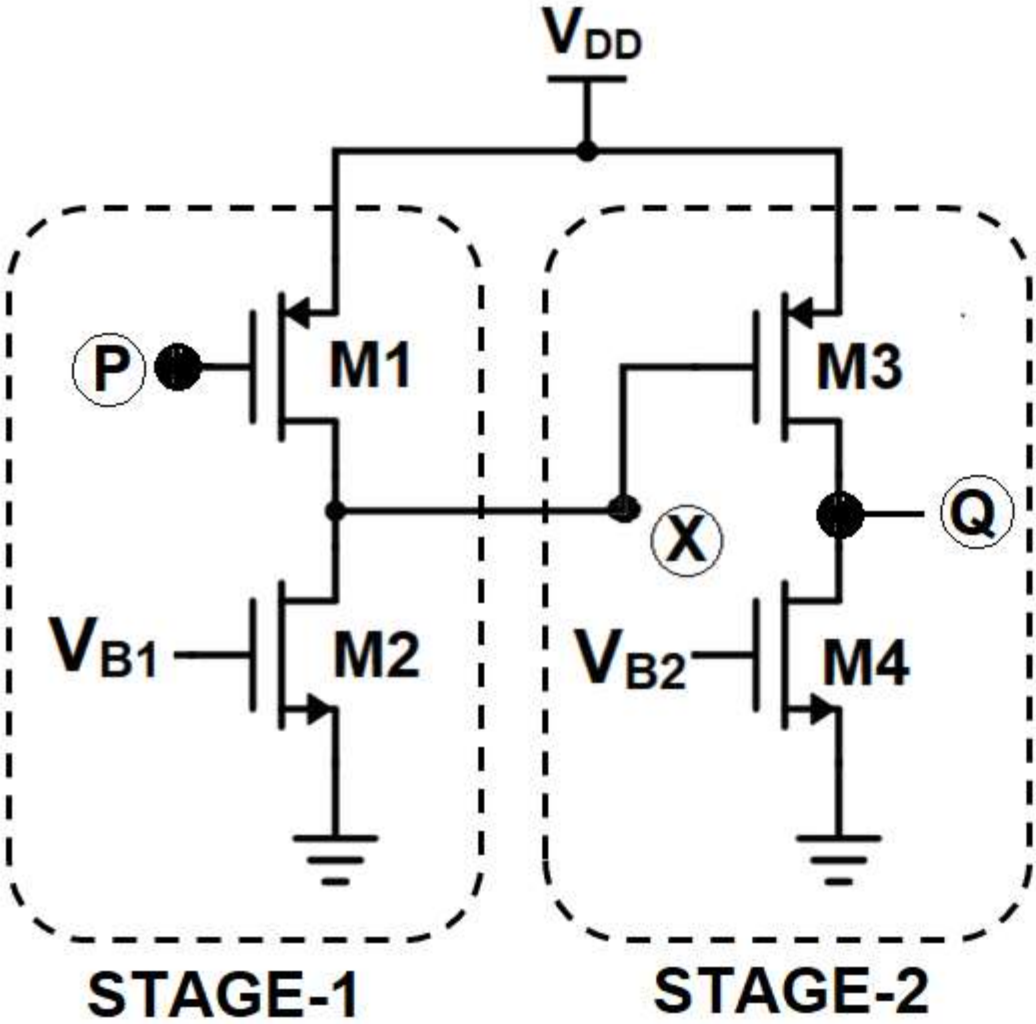
Prepared by

Dr. K. Babu Ravi Teja

Assistant Professor, Department of EEE

BITS Pilani, Pilani Campus

Mail: baburaviteja.k@pilani.bits-pilani.ac.in



V_DD
P
M1
M3
X
Q
V_B1
M2
V_B2
M4
STAGE-1
STAGE-2